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Dissolvable solid structure comprising first and second polymeric structurants

Abstract

A dissolvable solid structure that is formed by a homogeneous mixture. The mixture includes from about 1% to about 10% of a first polymeric structurant having a molecular weight of from about 1,000,000 g/mol to about 4,000,000 g/mol, from about 10% to about 20% of a second polymeric structurant having a molecular weight of from about 50,000 g/mol to about 500,000 g/mol, from about 10 wt % to about 85 wt % of one or more high melting point fatty material having a carbon chain length C12-C22 or mixtures thereof and a melting point above 25° C., and from about 1 wt % to about 60 wt % of a cationic surfactant. The dissolvable solid structures can provide an adequate strength during manufacturing, while not deteriorating usage feel when used and/or applied by consumers.

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D774086	12/2015	Montes et al.	N/A	N/A
D775198	12/2015	Montes et al.	N/A	N/A
9539444	12/2016	Kinoshita et al.	N/A	N/A
9545364	12/2016	Glenn, Jr. et al.	N/A	N/A
D778026	12/2016	Roetheli	N/A	N/A
D793025	12/2016	Slusarczyk et al.	N/A	N/A
D797551	12/2016	Chatterton et al.	N/A	N/A
D798143	12/2016	Chatterton	N/A	N/A
D808583	12/2017	Zietek	N/A	N/A
9902077	12/2017	Park et al.	N/A	N/A
D811922	12/2017	Lefave	N/A	N/A
D811935	12/2017	Hughes	N/A	N/A
D819836	12/2017	Noël	N/A	N/A
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D848102	12/2018	Carlson et al.	N/A	N/A
D850041	12/2018	Endle	N/A	N/A
10294586	12/2018	Sivik et al.	N/A	N/A
D851344	12/2018	Carlson et al.	N/A	N/A
D857156	12/2018	Hani	N/A	N/A
D857242	12/2018	Darrow et al.	N/A	N/A
D857929	12/2018	Darrow et al.	N/A	N/A
D862020	12/2018	Gorrell et al.	N/A	N/A
D863600	12/2018	Chao	N/A	N/A
D864507	12/2018	Stoughton et al.	N/A	N/A
D866105	12/2018	Carlson et al.	N/A	N/A
D866891	12/2018	Carlson et al.	N/A	N/A
D866892	12/2018	Hunt et al.	N/A	N/A
D866893	12/2018	Hunt et al.	N/A	N/A
D867717	12/2018	Chavez	N/A	N/A
D868159	12/2018	Swisher et al.	N/A	N/A
D868953	12/2018	Mckendree	N/A	N/A
10569286	12/2019	Anderson et al.	N/A	N/A
D878694	12/2019	Carlson et al.	N/A	N/A
10646413	12/2019	Sivik et al.	N/A	N/A
10694917	12/2019	Dreher et al.	N/A	N/A
D901115	12/2019	Carlson et al.	N/A	N/A
D903152	12/2019	Chao	N/A	N/A
10821056	12/2019	Swartz et al.	N/A	N/A
D906802	12/2020	Chi	N/A	N/A
10894005	12/2020	Sivik et al.	N/A	N/A
D910434	12/2020	Tan et al.	N/A	N/A
D910457	12/2020	Lee	N/A	N/A
D921166	12/2020	Meyers	N/A	N/A
D933095	12/2020	Heiner et al.	N/A	N/A
11351094	12/2021	Hamersky et al.	N/A	N/A
11395789	12/2021	Pratt et al.	N/A	N/A
11419808	12/2021	Hilvert et al.	N/A	N/A
11679066	12/2022	Song et al.	N/A	N/A

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2001/0037851	12/2000	Mortellite	N/A	N/A
2002/0018906	12/2001	Clark	N/A	N/A
2002/0044968	12/2001	Van Lengerich	N/A	N/A
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2002/0161088	12/2001	Kochvar	N/A	N/A
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2002/0173213	12/2001	Chu	N/A	N/A
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2003/0224959	12/2002	Smith	N/A	N/A
2003/0232183	12/2002	Dufton	N/A	N/A
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2004/0059055	12/2003	Inada et al.	N/A	N/A
2004/0071742	12/2003	Popplewell	N/A	N/A
2004/0071755	12/2003	Fox	N/A	N/A
2004/0082239	12/2003	Di Luccio et al.	N/A	N/A
2004/0092635	12/2003	Kitamura	N/A	N/A
2004/0108615	12/2003	Foley	N/A	N/A
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2004/0167256	12/2003	Verrall	N/A	N/A
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2004/0180597	12/2003	Kamada	N/A	N/A
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2007/0110792	12/2006	Simon	N/A	N/A
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2007/0134481	12/2006	Aubrun-sonneville	N/A	N/A
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2007/0253926	12/2006	Tadrowski	N/A	N/A
2007/0259170	12/2006	Brown	N/A	N/A
2007/0259996	12/2006	Vicari	N/A	N/A
2007/0269651	12/2006	Denome et al.	N/A	N/A
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2008/0019935	12/2007	Khan	N/A	N/A
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2008/0118727	12/2007	Andersen	N/A	N/A
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2008/0220054	12/2007	Shastri	N/A	N/A
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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a dissolvable solid structure, wherein the dissolvable solid structure is formed by a homogeneous mixture comprising, by weight of the structure: from about 1% to about 10% of a first polymeric structurant having a molecular weight of from about 1,000,000 g/mol to about 4,000,000 g/mol; from about 10% to about 20% of a second polymeric structurant having a molecular weight of from about 50,000 g/mol to about 500,000 g/mol; from about 10 wt % to about 85 wt % of one or more high melting point fatty material having a carbon chain length C12-C22 or mixtures thereof, wherein the melting point is above 25° C.; and from about 1 wt % to about 60 wt % of a cationic surfactant. The dissolvable solid structures of present invention provide an adequate strength during manufacturing, while not deteriorating usage feel when used and/or applied by consumers.

BACKGROUND OF THE INVENTION

(2) Many personal care and other consumer products in the market today are sold in liquid form. While widely used, liquid products often have tradeoffs in terms of packaging, storage, transportation, and convenience of use. Liquid consumer products typically are sold in bottles which add cost as well as packaging waste, much of which ends up in land-fills.

(3) Hair Care products in the form of a dissolvable solid structures present an attractive form to consumers. Market executions of dissolvable solid structures may include, dissolvable films, compressed powders in a solid, fibrous structures, porous foams, soluble deformable solids, powders, etc.

(4) For example, US Patent Application Publication No. 2018/0333339 A discloses a dissolvable solid structure which can be in the form of a fibrous structure comprising: (a) a polymeric structurant having a weight average molecular weight of from about 10,000 g/mol to about 6,000,000 g/mol; (b) a high melting point fatty compound such as a fatty amphiphile, and (c) a cationic surfactant.

(5) However, a need still exists for such dissolvable solid structures, especially for those containing a high melting point fatty compound and a cationic surfactant, which have an adequate strength during manufacturing, while not deteriorating usage feel when used and/or applied by consumers.

SUMMARY OF THE INVENTION

(6) The present invention is directed to a dissolvable solid structure, wherein the dissolvable solid structure is formed by a homogeneous mixture comprising, by weight of the structure: from about 1% to about 10% of a first polymeric structurant having a molecular weight of from about

1,000,000 g/mol to about 4,000,000 g/mol; from about 10% to about 20% of a second polymeric structurant having a molecular weight of from about 50,000 g/mol to about 500,000 g/mol; from about 10 wt % to about 85 wt % of one or more high melting point fatty material having a carbon chain length C12-C22 or mixtures thereof, wherein the melting point is above 25° C.; and from about 1 wt % to about 60 wt % of a cationic surfactant. (7) It has now been surprisingly found by the present inventors that, by the use of the first and second polymeric structurants having specific molecular weights, the dissolvable solid structures of present invention provide an adequate strength during manufacturing, while not deteriorating usage feel when used and/or applied by consumers.

Description

DETAILED DESCRIPTION OF THE INVENTION

Definitions

- (1) As used herein, The Dissolvable Solid Structure may be referred to herein as “the Dissolvable Solid Structure”, “the Structure”, or “the Dissolvable Structure”.
- (2) As used herein, “dissolvable” means that the Dissolvable Solid Structure is completely soluble in water or it provides a uniform dispersion upon mixing in water according to the hand dissolution test. The Dissolvable Solid Structure has a hand dissolution value of from about 1 to about 30 strokes, alternatively from about 2 to about 25 strokes, alternatively from about 3 to about 20 strokes, and alternatively from about 4 to about 15 strokes, as measured by the Hand Dissolution Method.
- (3) As used herein, “flexible” means a Dissolvable Solid Structure meets the distance to maximum force values discussed herein.
- (4) “Fibrous structure” as used herein means a structure that comprises one or more fibrous elements and optionally, one or more particles. The fibrous structure as described herein can mean an association of fibrous elements and optionally, particles that together form a structure, such as a unitary structure, capable of performing a function.
- (5) The fibrous structure may have a multi-ply fibrous structure which comprises two or more different fibrous structure plies. Each ply may be the same as or different from other ply.
- (6) A layer comprising fibrous elements may sometimes be referred to as a ply. A ply may be a fibrous structure which may be homogeneous or layered as described herein.
- (7) The single-ply fibrous structure or a multi-ply fibrous structure comprising one or more fibrous structure plies as described herein may exhibit a basis weight of less than 5000 g/m.sup.2 as measured according to the Basis Weight Test Method described herein. For example, the single- or multi-ply fibrous structure may exhibit a basis weight of greater than 10 g/m.sup.2 to about 5000 g/m.sup.2 and/or greater than 10 g/m.sup.2 to about 3000 g/m.sup.2 and/or greater than 10 g/m.sup.2 to about 2000 g/m.sup.2 and/or greater than 10 g/m.sup.2 to about 1000 g/m.sup.2 and/or greater than 20 g/m.sup.2 to about 800 g/m.sup.2 and/or greater than 30 g/m.sup.2 to about 600 g/m.sup.2 and/or greater than 50 g/m.sup.2 to about 500 g/m.sup.2 and/or greater than 300 g/m.sup.2 to about 3000 g/m.sup.2 and/or greater than 500 g/m.sup.2 to about 2000 g/m.sup.2 as measured according to the Basis Weight Test Method.
- (8) In one example, the fibrous structure can be a “unitary fibrous structure.”
- (9) “Unitary fibrous structure” as used herein is an arrangement comprising a plurality of two or more and/or three or more fibrous elements that are inter-entangled or otherwise associated with one another to form a fibrous structure and/or fibrous structure plies. A unitary fibrous structure may be one or more plies within a multi-ply fibrous structure. In one example, a unitary fibrous structure may comprise three or more different fibrous elements. In another example, a unitary fibrous structure may comprise two or more different fibrous elements.
- (10) “Article” as used herein refers to a consumer use unit, a consumer unit dose unit, a consumer use saleable unit, a single dose unit, or other use form comprising a unitary fibrous dissolvable solid structure and/or comprising one or more fibrous structures.
- (11) “Fibrous element” as used herein means an elongated particulate having a length greatly exceeding its average diameter, i.e., a length to average diameter ratio of at least about 10. A fibrous element may be a filament or a fiber. In one example, the fibrous element is a single fibrous element rather than a yarn comprising a plurality of fibrous elements.
- (12) The fibrous elements may be spun from a filament-forming composition also referred to as fibrous element-forming compositions via suitable spinning process operations, such as meltblowing, spunbonding, electro-spinning, and/or rotary spinning.
- (13) The fibrous elements may be monocomponent (single, unitary solid piece rather than two different parts, like a core/sheath bicomponent) and/or multicomponent. For example, the fibrous elements may comprise bicomponent fibers and/or filaments. The bicomponent fibers and/or filaments may be in any form, such as side-by-side, core and sheath, islands-in-the-sea and the like.
- (14) “Filament” as used herein means an elongated particulate as described above that exhibits a length of greater than or equal to 5.08 cm (2 in.) and/or greater than or equal to 7.62 cm (3 in.) and/or greater than or equal to 10.16 cm (4 in.) and/or greater than or equal to 15.24 cm (6 in.). Filaments are typically considered continuous or substantially continuous in nature. Filaments are relatively longer than fibers. Non-limiting examples of filaments include meltblown and/or spunbond filaments.
- (15) “Fiber” as used herein means an elongated particulate as described above that exhibits a length of less than 5.08 cm (2 in.) and/or less than 3.81 cm (1.5 in.) and/or less than 2.54 cm (1 in.). Fibers are typically considered discontinuous in nature. Non-limiting examples of fibers include staple fibers produced by spinning a filament or filament tow and then cutting the filament or filament tow into segments of less than 5.08 cm (2 in.) thus producing fibers. Therefore, references to filaments herein also include fibers made from such filaments unless otherwise noted. Fibers are typically considered discontinuous in nature relative to filaments, which are considered continuous in nature.
- (16) “Filament-forming composition” and/or “fibrous element-forming composition” as used herein means a composition that is suitable for making a fibrous element of the present invention such as by meltblowing and/or spunbonding. The filament-forming composition comprises one or more filament-forming materials that exhibit properties that make them suitable for spinning into a fibrous element. In one example, the filament-forming material comprises a polymer. In addition to one or more filament-forming materials, the filament-forming composition may comprise one or more additives, for example one or more active agents. In addition, the filament-forming composition may comprise one or more polar solvents, such as water, into which one or more, for example all, of the filament-forming materials and/or one or more, for example all, of the active agents are dissolved and/or dispersed prior to spinning a fibrous element, such as a filament from the filament-forming composition.
- (17) As used herein, “porous” means that the Dissolvable Solid Structure has spaces, voids or interstices, (generally referred to herein as “pores”) provided by the microscopic complex three-dimensional configuration, that provide channels, paths or passages through which a liquid can flow.
- (18) As used herein, “porosity” and “percent porosity” are used interchangeably and each refers to a measure of void volume of the Dissolvable Solid Structure and is calculated as
$$[1 - (\text{basis weight of Dissolvable Solid Structure} / [\text{thickness of Dissolvable Solid Structure} \times \text{density of the bulk, dried material}])] \times 100\%$$
 with the units adjusted so they cancel and multiplied by 100% to provide percent porosity.
- (19) The Dissolvable Solid Structure may be referred to herein as “the Dissolvable Solid Structure” or “the Dissolvable Structure”.
- (20) The term “molecular weight” or “Molecular weight” refers to the weight average molecular weight unless otherwise stated. Molecular weight is measured using industry standard method, gel permeation chromatography (“GPC”).
- (21) As used herein, the articles including “a” and “an” when used in a claim, are understood to mean one or more of what is claimed or described.
- (22) As used herein, the terms “include,” “includes,” and “including,” are meant to be non-limiting.
- (23) The methods disclosed in the Test Methods Section of the present application should be used to determine the respective values of the

parameters of Applicants' inventions, including those discussed in the Dissolvable Structures—Physical Characteristics section below.

(24) All percentages and ratios are calculated by weight unless otherwise indicated. All percentages and ratios are calculated based on the total composition unless otherwise indicated.

(25) It should be understood that every maximum numerical limitation given throughout this specification includes every lower numerical limitation, as if such lower numerical limitations were expressly written herein. Every minimum numerical limitation given throughout this specification will include every higher numerical limitation, as if such higher numerical limitations were expressly written herein. Every numerical range given throughout this specification will include every narrower numerical range that falls within such broader numerical range, as if such narrower numerical ranges were all expressly written herein.

(26) Dissolvable Solid Structure (Dissolvable Solid Articles)

(27) The dissolvable solid structure of the present invention is formed by a homogeneous mixture, comprises, by weight of the structure: from about 1% to about 10% of a first polymeric structurant having a molecular weight of from about 1,000,000 g/mol to about 4,000,000 g/mol; from about 10% to about 20% of a second polymeric structurant having a molecular weight of from about 50,000 g/mol to about 500,000 g/mol; from about 10 wt % to about 85 wt % of one or more high melting point fatty material having a carbon chain length C12-C22 or mixtures thereof, wherein the melting point is above 25° C.; and from about 1 wt % to about 60 wt % of a cationic surfactant.

(28) The dissolvable solid structure can be a fibrous structure formed by a plurality of fibrous elements wherein the fibrous elements are formed by the above homogeneous mixture.

(29) The dissolvable solid structure of the present invention can be a personal care product, for example a hair care product, alternatively rinse-off hair care product, even alternatively rinse-off hair care product containing non-sulfate surfactant.

(30) The dissolvable solid structure may contain particulates. The dissolvable solid structure may have an aesthetic feature selected from the group consisting of printing, embossing, texture, colored and mixtures thereof.

(31) First and Second Polymeric Structurants

(32) The structure comprises: from about 1% to about 10% of a first polymeric structurant having a molecular weight of from about 1,000,000 g/mol to about 4,000,000 g/mol, alternatively from about 1,500,000 g/mol to about 3,500,000 g/mol; and from about 10% to about 20% of a second polymeric structurant having a molecular weight of from about 50,000 g/mol to about 500,000 g/mol, alternatively from about 65,000 g/mol to about 395,000 g/mol.

(33) In the present invention, it is preferred the weight ratio between the first and second polyvinylpyrrolidones is from about structurants is from about 1:1 to about 1:100, alternatively from about 1:1 to about 1:50, alternatively from about 1:1 to about 1:10.

(34) In the present invention, it is preferred that the weight ratio between the first and second polymeric structurant is in a range such that the combined average molecular weight of the first and second polymeric structurants is from about 300,000 g/mol to about 3,000,000 g/mol, alternatively from about 350,000 g/mol to about 2,000,000 g/mol, alternatively from about 400,000 g/mol to about 1,000,000 g/mol, and alternatively from about 500,000 g/mol to about 1,000,000 g/mol.

(35) In the present invention, it is preferred that the amounts of the first and second polymeric structurants, the molecular weights of the first and second polymeric structurants, and the weight ratio between the first and second polymeric structurants are selected such that the mixture in melt at 85° C. has a viscosity of from about 0.3 Pa·s to about 10 Pa·Math.s, alternatively from about 3 Pa·Math.s to about 8 Pa·Math.s.

(36) The first and second polymeric structurants are water-soluble polymers. As used in the present invention, the term “water-soluble polymer” is broad enough to include both water-soluble and water-dispersible polymers, and is defined as a polymer with a solubility in water, measured at 25° C., of at least about 0.1 gram/liter (g/L). In some embodiments, the polymers have a solubility in water, measured at 25° C., of from about 0.1 gram/liter (g/L) to about 500 grams/liter (g/L). (This indicates production of a macroscopically isotropic or transparent, colored or colorless solution). The polymers for making these solids may be of synthetic or natural origin and may be modified by means of chemical reactions. They may or may not be film-forming. These polymers should be physiologically acceptable, i.e., they should be compatible with the skin, mucous membranes, the hair and the scalp. The terms “water-soluble polymer” and “polymer structurant” are used interchangeably herein. Furthermore, whenever the singular term “polymer” is stated, it should be understood that the term is broad enough to include one polymer or a mixture of more than one polymer. For instance, if a mixture of polymers is used, the polymer solubility as referred to herein would refer to the solubility of the mixture of polymers, rather than to the solubility of each polymer individually.

(37) Non-limiting examples of such water-soluble polymers can include polyacrylic acid and its copolymers, polyvinylpyrrolidone and its copolymers, polyacrylamide and its copolymers, polyvinylmethyl ether, polyethyleneimine, polymethacrylic acid, other water soluble acrylic polymers such as polyN-isopropyl acrylamide, polyN-N-dimethylacrylamide, poly (2-ethyl-2-oxazoline), polyoxazoline, polyvinylloxazolidone, polyvinyl caprolactam, polystyrene sulfonate, polyvinyl formamide, polyvinyl amine, alkylated polyvinyl pyrrolidone, poly vinyl caprolactam, polyvinyl valerolactam, polyvinyl imidazole, polyacrylic acid, polyacrylamide, polymethacrylamide, polydimethacrylamide, polyalkylaminomethacrylate, and polyalkylaminomethacrylamide, and combinations thereof. For purposes of clarity, the use of the term “copolymer” is intended to convey that the vinyl pyrrolidone monomer can be copolymerized with other non-limiting monomers such as vinyl acetate, alkylated vinyl pyrrolidone, vinyl caprolactam, vinyl valerolactam, vinyl imidazole, acrylic acid, methacrylate, acrylamide, vinyl oxazoline, 2-ethyl-2oxazoline, methacrylamide, dimethacrylamide, alkylaminomethacrylate, and alkylaminomethacrylamide monomers.

(38) Among a variety of the water soluble polymers, the first and second polymeric structurants can be selected from the group consisting of: polyvinylpyrrolidone and its copolymers, and mixtures thereof. alternatively, the first and second polymeric structurants are polyvinylpyrrolidone.

(39) As used herein, “vinyl pyrrolidone copolymer” (and “copolymer” when used in reference thereto) refers to a polymer of the following structure (I): (I)

(40) ##STR00001##

In structure (I), n is an integer such that the polymeric structurant has the degree of polymerization such that it possesses characteristics described herein. For purposes of clarity, the use of the term “copolymer” is intended to convey that the vinyl pyrrolidone monomer can be copolymerized with other non-limiting monomers such as vinyl acetate, alkylated vinyl pyrrolidone, vinyl caprolactam, vinyl valerolactam, vinyl oxazoline, 2-ethyl-2oxazoline, vinyl imidazole, acrylic acid, methacrylate, acrylamide, methacrylamide, dimethacrylamide, alkylaminomethacrylate, and alkylaminomethacrylamide monomers.

(41) For example, suitable polymers for use are PVP K120 from Ashland having a weight average molecular weight of about 3,470,000 g/mol, PVP K90 from Ashland having a weight average molecular weight of 1,570,000 g/mol, PVP K60 from Jarchem having a weight average molecular weight of 396,000 g/mol, PVP K30 from Ashland having a weight average molecular weight of 66,800 g/mol. Additional suitable polymers include copolymers of polyvinylpyrrolidone, such as Ganex® or PVP/VA (weight average molecular weight of about 50,000 g/mol) copolymers from Ashland Inc., also performed as suitable structurants but a higher level was utilized to be effective due to their lower weight average molecular weight.

(42) Additional Polymeric Structurant

(43) The structure may further contain an additional structurant. Such additional structurant can be the polymers listed above, but those having different MW from FIRST and SECOND

(44) Polymer Structurants.

(45) Cationic Surfactant

(46) The structure further comprises a cationic surfactant at a level of from about 1 wt % to about 60 wt %, alternatively from about 10 wt % to about

50 wt % to alternatively from about 20 wt % to about 40 wt % by weight of the structure.

(47) Cationic surfactant useful herein can be one cationic surfactant or a mixture of two or more cationic surfactants. The cationic surfactant can be selected from the group consisting of, but not limited to: a mono-long alkyl quaternized ammonium salt; a combination of a mono-long alkyl quaternized ammonium salt and a di-long alkyl quaternized ammonium salt; a mono-long alkyl amine; a combination of a mono-long alkyl amine and a di-long alkyl quaternized ammonium salt; and a combination of a mono-long alkyl amine and a mono-long alkyl quaternized ammonium salt, a tertiary amine and combinations thereof.

(48) Mono-Long Alkyl Amine

(49) Mono-long alkyl amine useful herein are those having one long alkyl chain of from 12 to 30 carbon atoms, alternatively from 16 to 24 carbon atoms, alternatively from 18 to 22 alkyl group. Mono-long alkyl amines useful herein also include mono-long alkyl amidoamines. Primary, secondary, and tertiary fatty amines are useful.

(50) Suitable for use in the dissolvable solid structure are tertiary amido amines having an alkyl group of from about 12 to about 22 carbons.

Exemplary tertiary amido amines include: stearamidopropyldimethylamine, stearamidopropyldiethylamine, stearamidoethyldiethylamine, stearamidoethyldimethylamine, palmitamidopropyldimethylamine, palmitamidopropyldiethylamine, palmitamidoethyldiethylamine, palmitamidoethyldimethylamine, behenamidopropyldimethylamine, behenamidopropyldiethylamine, behenamidoethyldiethylamine, behenamidoethyldimethylamine, arachidamidopropyldimethylamine, arachidamidopropyldiethylamine, arachidamidoethyldiethylamine, arachidamidoethyldimethylamine, diethylaminoethylstearamide. Useful amines in the present invention are disclosed in U.S. Pat. No. 4,275,055, Nachtigal, et al.

(51) These amines can be used in combination with acids such as 1-glutamic acid, lactic acid, hydrochloric acid, malic acid, succinic acid, acetic acid, fumaric acid, tartaric acid, citric acid, 1-glutamic hydrochloride, maleic acid, and mixtures thereof; alternatively 1-glutamic acid, lactic acid, citric acid, at a molar ratio of the amine to the acid of from about 1:0.3 to about 1:2, alternatively from about 1:0.4 to about 1:1.

(52) Mono-Long Alkyl Quaternized Ammonium Salt

(53) The mono-long alkyl quaternized ammonium salts useful herein are those having one long alkyl chain which has from 12 to 30 carbon atoms, alternatively from 16 to 24 carbon atoms, alternatively a C18-22 alkyl group. The remaining groups attached to nitrogen are independently selected from an alkyl group of from 1 to about 4 carbon atoms or an alkoxy, polyoxyalkylene, alkylamido, hydroxyalkyl, aryl or alkylaryl group having up to about 4 carbon atoms.

(54) Mono-long alkyl quaternized ammonium salts useful herein are those having the formula (I):

(55) ##STR00002##

wherein one of R.sup.75, R.sup.76, R.sup.77 and R.sup.78 is selected from an alkyl group of from 12 to 30 carbon atoms or an aromatic, alkoxy, polyoxyalkylene, alkylamido, hydroxyalkyl, aryl or alkylaryl group having up to about 30 carbon atoms; the remainder of R.sup.75, R.sup.76, R.sup.77 and R.sup.78 are independently selected from an alkyl group of from 1 to about 4 carbon atoms or an alkoxy, polyoxyalkylene, alkylamido, hydroxyalkyl, aryl or alkylaryl group having up to about 4 carbon atoms; and X.sup.- is a salt-forming anion such as those selected from halogen, (e.g. chloride, bromide), acetate, citrate, lactate, glycolate, phosphate, nitrate, sulfonate, sulfate, alkylsulfate, and alkyl sulfonate radicals. The alkyl groups can contain, in addition to carbon and hydrogen atoms, ether and/or ester linkages, and other groups such as amino groups. The longer chain alkyl groups, e.g., those of about 12 carbons, or higher, can be saturated or unsaturated. One of R.sup.75, R.sup.76, R.sup.77 and R.sup.78 can be selected from an alkyl group of from 12 to 30 carbon atoms, alternatively from 16 to 24 carbon atoms, alternatively from 18 to 22 carbon atoms, alternatively 22 carbon atoms; the remainder of R.sup.75, R.sup.76, R.sup.77 and R.sup.78 can be independently selected from CH.sub.3, C.sub.2H.sub.5, C.sub.2H.sub.4OH, and mixtures thereof; and X can be selected from the group consisting of Cl, Br, CH.sub.3OSO.sub.3, C.sub.2H.sub.5OSO.sub.3, and mixtures thereof.

(56) Nonlimiting examples of such mono-long alkyl quaternized ammonium salt cationic surfactants include: behenyl trimethyl ammonium salt; stearyl trimethyl ammonium salt; cetyl trimethyl ammonium salt; and hydrogenated tallow alkyl trimethyl ammonium salt.

(57) Di-Long Alkyl Quaternized Ammonium Salts

(58) When used, di-long alkyl quaternized ammonium salts can be combined with a mono-long alkyl quaternized ammonium salt and/or mono-long alkyl amine salt, at the weight ratio of from 1:1 to 1:5, alternatively from 1:1.2 to 1:5, alternatively from 1:1.5 to 1:4, in view of stability in rheology and conditioning benefits.

(59) Di-long alkyl quaternized ammonium salts useful herein are those having two long alkyl chains of from 12 to 30 carbon atoms, alternatively from 16 to 24 carbon atoms, alternatively from 18 to 22 carbon atoms. Such di-long alkyl quaternized ammonium salts useful herein are those having the formula (I):

(60) ##STR00003##

wherein two of R.sup.71, R.sup.72, R.sup.73 and R.sup.74 are selected from an aliphatic group of from 12 to 30 carbon atoms, alternatively from 16 to 24 carbon atoms, alternatively from 18 to 22 carbon atoms or an aromatic, alkoxy, polyoxyalkylene, alkylamido, hydroxyalkyl, aryl or alkylaryl group having up to about 30 carbon atoms; the remainder of R.sup.71, R.sup.72, R.sup.73 and R.sup.74 are independently selected from an aliphatic group of from 1 to about 8 carbon atoms, alternatively from 1 to 3 carbon atoms or an aromatic, alkoxy, polyoxyalkylene, alkylamido, hydroxyalkyl, aryl or alkylaryl group having up to about 8 carbon atoms; and X.sup.- is a salt-forming anion selected from the group consisting of halides such as chloride and bromide, C.sub.1-C.sub.4 alkyl sulfate such as methosulfate and ethosulfate, and mixtures thereof. The aliphatic groups can contain, in addition to carbon and hydrogen atoms, ether linkages, and other groups such as amino groups. The longer chain aliphatic groups, e.g., those of about 16 carbons, or higher, can be saturated or unsaturated. Two of R.sup.71, R.sup.72, R.sup.73 and R.sup.74 can be selected from an alkyl group of from 12 to 30 carbon atoms, alternatively from 16 to 24 carbon atoms, alternatively from 18 to 22 carbon atoms; and the remainder of R.sup.71, R.sup.72, R.sup.73 and R.sup.74 are independently selected from CH.sub.3, C.sub.2H.sub.5, C.sub.2H.sub.4OH, CH.sub.2C.sub.6H.sub.5, and mixtures thereof.

(61) Suitable di-long alkyl cationic surfactants include, for example, dialkyl (14-18) dimethyl ammonium chloride, ditallow alkyl dimethyl ammonium chloride, dihydrogenated tallow alkyl dimethyl ammonium chloride, distearyl dimethyl ammonium chloride, and dicetyl dimethyl ammonium chloride.

(62) High Melting Point Fatty Compound

(63) The structure further comprises a high melting point fatty compound at a level of from about 10 wt % to about 85 wt %, alternatively from 20 wt % to 70 wt %, alternatively from about 50 wt % to about 70 wt %, alternatively from about 10 wt % to about 20 wt % by weight of the structure. The fatty compound can be selected from the group consisting of, but not limited to, fatty amphiphiles, fatty alcohol, fatty acid, fatty amide, fatty ester and combinations thereof.

(64) The high melting point fatty compound useful herein have a melting point of 25° C. or higher, alternatively 40° C. or higher, alternatively 45° C. or higher, alternatively 50° C. or higher, in view of stability of the emulsion especially the gel matrix. Such melting point is up to about 90° C., alternatively up to about 80° C., alternatively up to about 70° C., alternatively up to about 65° C., in view of easier manufacturing and easier emulsification. The high melting point fatty compound can be used as a single compound or as a blend or mixture of at least two high melting point fatty compounds. When used as such blend or mixture, the above melting point means the melting point of the blend or mixture.

(65) The high melting point fatty compound useful herein is selected from the group consisting of fatty alcohols, fatty acids, fatty alcohol derivatives, fatty acid derivatives, fatty amides, and mixtures thereof. It is understood by the artisan that the compounds disclosed in this section of the specification can in some instances fall into more than one classification, e.g., some fatty alcohol derivatives can also be classified as fatty acid

derivatives. However, a given classification is not intended to be a limitation on that particular compound, but is done so for convenience of classification and nomenclature. Further, it is understood by the artisan that, depending on the number and position of double bonds, and length and position of the branches, certain compounds having certain required carbon atoms may have a melting point of less than the above. Such compounds of low melting point are not intended to be included in this section. Nonlimiting examples of the high melting point compounds are found in International Cosmetic Ingredient Dictionary, Fifth Edition, 1993, and CTFA Cosmetic Ingredient Handbook, Second Edition, 1992.

(66) Among a variety of high melting point fatty compounds, fatty alcohols can be used in the composition described herein. The fatty alcohols useful herein are those having from about 14 to about 30 carbon atoms, alternatively from about 16 to about 22 carbon atoms. These fatty alcohols are saturated and can be straight or branched chain alcohols.

(67) Suitable fatty alcohols include, but are not limited to, cetyl alcohol (having a melting point of about 56° C.), stearyl alcohol (having a melting point of about 58-59° C.), behenyl alcohol (having a melting point of about 71° C.), and mixtures thereof. These compounds are known to have the above melting point. However, they often have lower melting points when supplied, since such supplied products are often mixtures of fatty alcohols having alkyl chain length distribution in which the main alkyl chain is cetyl, stearyl or behenyl group.

(68) Generally, in the mixture, the weight ratio of cetyl alcohol to stearyl alcohol is from about 1:9 to 9:1, alternatively from about 1:4 to about 4:1, alternatively from about 1:2.3 to about 1.5:1.

(69) When using higher level of total cationic surfactant and high melting point fatty compounds, the mixture has the weight ratio of cetyl alcohol to stearyl alcohol of from about 1:1 to about 4:1, alternatively from about 1:1 to about 2:1, alternatively from about 1.2:1 to about 2:1, in view of maintaining acceptable consumer usage. It may also provide more conditioning on damaged part of the hair.

(70) Dispersing Agent

(71) The structure may further comprise a dispersing agent for increasing the wetting, hydration, and/or dispersion of the conditioner materials. The dispersing agent can be included at a level of from about 1 wt % to about 30 wt %, alternatively from about 5 wt % to about 15 wt %, and alternatively from about 5 wt % to about 10 wt % by weight of the second layer. A surfactant from the nonionic class of alkyl glucamides can improve the wetting and hydration when added to the solid conditioner formula. The alkyl glucamide surfactant contains a hydrophobic tail of about 8-18 carbons and a nonionic head group of glucamide. For glucamide, the presence of the amide and hydroxyl groups may provide sufficient polarity that balances the hydrophobic carbon tail in such a way to permit the surfactant's solubility in the conditioner oils and also imparts a rapid dispersion of the conditioner ingredients upon exposure to water. Other similar dispersing agents include, but are not limited to, reverse alkyl glucamides, gluconamides, cocoamidopropyl betaines, alkyl glucoside, triethanol amine, cocamide MEAs and mixtures thereof.

(72) Plasticizer

(73) The structure may optionally comprise from about 1 wt % to about 25 wt % plasticizer, in one embodiment from about 3 wt % to about 20 wt % plasticizer, in one embodiment from about 5 wt % to about 15 wt % plasticizer by weight of the layer.

(74) When present in the Structures, non-limiting examples of suitable plasticizing agents include polyols, copolyols, polycarboxylic acids, polyesters and dimethicone copolyols.

(75) Examples of useful polyols include, but are not limited to, glycerin, diglycerin, propylene glycol, ethylene glycol, butylene glycol, pentylene glycol, cyclohexane dimethanol, hexane diol, polyethylene glycol (200-600), sugar alcohols such as sorbitol, manitol, lactitol, isosorbide, glucamine, N-methylglucamine and other mono- and polyhydric low molecular weight alcohols (e.g., C.sub.2-C.sub.8 alcohols); mono di- and oligo-saccharides such as fructose, glucose, sucrose, maltose, lactose, and high fructose corn syrup solids and ascorbic acid.

(76) Examples of polycarboxylic acids include, but are not limited to citric acid, maleic acid, succinic acid, polyacrylic acid, and polymaleic acid.

(77) Examples of suitable polyesters include, but are not limited to, glycerol triacetate, acetylated-monoglyceride, diethyl phthalate, triethyl citrate, tributyl citrate, acetyl triethyl citrate, acetyl tributyl citrate.

(78) Examples of suitable dimethicone copolyols include, but are not limited to, PEG-12 dimethicone, PEG/PPG-18/18 dimethicone, and PPG-12 dimethicone.

(79) Other suitable plasticizers include, but are not limited to, alkyl and allyl phthalates; naphthalates; lactates (e.g., sodium, ammonium and potassium salts); sorbeth-30; urea; lactic acid; sodium pyrrolidone carboxylic acid (PCA); sodium hyaluronate or hyaluronic acid; soluble collagen; modified protein; monosodium L-glutamate; alpha & beta hydroxyl acids such as glycolic acid, lactic acid, citric acid, maleic acid and salicylic acid; glyceryl polymethacrylate; polymeric plasticizers such as polyquaterniums; proteins and amino acids such as glutamic acid, aspartic acid, and lysine; hydrogen starch hydrolysates; other low molecular weight esters (e.g., esters of C.sub.2-C.sub.10 alcohols and acids); and any other water soluble plasticizer known to one skilled in the art of the foods and plastics industries; and mixtures thereof.

(80) EP 0283165 B1 discloses suitable plasticizers, including glycerol derivatives such as propoxylated glycerol.

(81) Optional Ingredients

(82) The first and/or the second layer may comprise other optional ingredients that are known for use or otherwise useful in compositions, provided that such optional materials are compatible with the selected essential materials described herein, or do not otherwise unduly impair product performance.

(83) Such optional ingredients are most typically those materials approved for use in cosmetics and that are described in reference books such as the CTFA Cosmetic Ingredient Handbook, Second Edition, The Cosmetic, Toiletries, and Fragrance Association, Inc. 1992.

(84) Further non-limiting examples of such optional ingredients include preservatives, perfumes or fragrances, coloring agents or dyes, conditioning agents, hair bleaching agents, thickeners, moisturizers, emollients, pharmaceutical actives, vitamins or nutrients, sunscreens, deodorants, sensates, plant extracts, nutrients, astringents, cosmetic particles, absorbent particles, adhesive particles, hair fixatives, fibers, reactive agents, skin lightening agents, skin tanning agents, anti-dandruff agents, perfumes, exfoliating agents, acids, bases, humectants, enzymes, suspending agents, pH modifiers, hair colorants, hair perming agents, pigment particles, anti-acne agents, anti-microbial agents, sunscreens, tanning agents, exfoliation particles, hair growth or restorer agents, insect repellents, shaving lotion agents, co-solvents or other additional solvents, and similar other materials. Further non-limiting examples of optional ingredients include encapsulated perfumes, such as by β -cyclodextrins, polymer microcapsules, starch encapsulated accords and combinations thereof.

(85) Suitable conditioning agents include high melting point fatty compounds, silicone conditioning agents and cationic conditioning polymers. Suitable materials are discussed in US 2008/0019935, US 2008/0242584 and US 2006/0217288.

(86) Physical Properties of the Dissolvable Solid Structure

(87) For fibrous Structures, the Structure comprises a significant number of dissolvable fibers with an average diameter less than about 150 micron, alternatively less than about 100 micron, alternatively less than about 10 micron, and alternatively less than about 1 micron with a relative standard deviation of less than 100%, alternatively less than 80%, alternatively less than 60%, alternatively less than 50%, such as in the range of 10% to 50%, for example. As set forth herein, the significant number means at least 10% of all the dissolvable fibers, alternatively at least 25% of all the dissolvable fibers, alternatively at least 50% of all the dissolvable fibers, alternatively at least 75% of all the dissolvable fibers. The significant number may be at least 99% of all the dissolvable fibers. Alternatively, from about 50% to about 100% of all the dissolvable fibers may have an average diameter less than about 10 micron. The dissolvable fibers produced by the method of the present disclosure have a significant number of dissolvable fibers with an average diameter less than about 1 micron, or sub-micron fibers. In an embodiment, Dissolvable Solid Structure may have from about 25% to about 100% of all the dissolvable fibers with an average diameter less than about 1 micron, alternatively from about 35% to about 100% of all the dissolvable fibers with an average diameter less than about 1 micron, alternatively from about 50% to about 100% of all the dissolvable fibers with an average diameter less than about 1 micron, and alternatively from about 75% to about 100% of all the dissolvable fibers

with an average diameter less than about 1 micron.

(88) The percent porosity of the dissolvable solid Structure is at least about 25%, alternatively at embodiment at least about 50%, alternatively at least about 60%, alternatively at least about 70% and alternatively at least about 80%. The porosity of the dissolvable solid Structure is not more than about 99%, alternatively not more than about 98%, alternatively not more than about 95%, and alternatively not more than about 90%. Porosity of a Structure is determined according to the procedure set forth in the definition of “porosity” above.

(89) A range of effective sizes of pores can be accommodated. The pore size distribution through the Structure cross-section may be symmetric or asymmetric.

(90) The Structure can be flexible and have a distance to maximum force value of from about 6 mm to about 30 mm. The distance to maximum force value from about 7 mm to about 25 mm, alternatively from about 8 mm to about 20 mm, and alternatively from about 9 mm to about 15 mm.

(91) The Structure can be characterized in one aspect by its Specific Surface Area. The Structure can have a Specific Surface Area of from about 0.03 m.sup.2/g to about 0.25 m.sup.2/g, alternatively from about 0.035 m.sup.2/g to about 0.22 m.sup.2/g, alternatively from about 0.04 m.sup.2/g to about 0.19 m.sup.2/g, and alternatively from about 0.045 m.sup.2/g to about 0.16 m.sup.2/g.

(92) The Structure can be a flat, flexible Structure in the form of a pad, a strip, or tape and having a thickness of from about 0.5 mm to about 10 mm, alternatively from about 1 mm to about 9 mm, alternatively from about 2 mm to about 8 mm, and alternatively from about 3 mm to about 7 mm as measured by the below methodology. The Structure can be a sheet having a thickness from about 5 mm to about 6.5 mm. Alternatively, two or more sheets are combined to form a Structure with a thickness of about 5 mm to about 10 mm.

(93) The Structure can have a basis weight of from about 200 grams/m.sup.2 to about 2,000 grams/m.sup.2, alternatively from about 400 g/m.sup.2 to about 1,200 g/m.sup.2, alternatively from about 600 g/m.sup.2 to about 2,000 g/m.sup.2, and alternatively from about 700 g/m.sup.2 to about 1,500 g/m.sup.2.

(94) The Structure can have a dry density of from about 0.08 g/cm.sup.3 to about 0.40 g/cm.sup.3, alternatively from about 0.08 g/cm.sup.3 to about 0.38 g/cm.sup.3, alternatively from about 0.10 g/cm.sup.3 to about 0.25 g/cm.sup.3, and alternatively from about 0.12 g/cm.sup.3 to about 0.20 g/cm.sup.3.

(95) Method of Use

(96) The dissolvable solid substrates described herein may be used for cleaning and/or treating hair, hair follicles, skin, teeth, and the oral cavity. The method for treating these consumer substrates may comprise the steps of: a) applying an effective amount of the Structure to the hand, b) wetting the Structure with water to dissolve the solid, c) applying the dissolved material to the target consumer substrate such as to clean or treat it, and d) rinsing the diluted treatment composition from the consumer substrate. These steps can be repeated as many times as desired to achieve the desired cleansing and or treatment benefit.

(97) A method useful for providing a benefit to hair, hair follicles, skin, teeth, and/or the oral cavity, includes the step of applying a composition according to the first embodiment to these target consumer substrates in need of regulating.

(98) Alternatively, a useful method for regulating the condition of hair, hair follicles, skin, teeth, the oral cavity, includes the step of applying one or more compositions described herein to these target consumer substrates in need of regulation.

(99) The amount of the composition applied, the frequency of application and the period of use will vary widely depending upon the purpose of application, the level of components of a given composition and the level of regulation desired. For example, when the composition is applied for whole body or hair treatment, effective amounts generally range from about 0.5 grams to about 10 grams, alternatively from about 1.0 grams to about 5 grams, and alternatively from about 1.5 grams to about 3 grams.

(100) Product Types and Articles of Commerce

(101) Non-limiting examples of products that utilize the dissolvable solid structures include hand cleansing substrates, teeth cleaning or treating substrates, oral cavity substrates, hair shampoo or other hair treatment substrates, body cleansing substrates, shaving preparation substrates, personal care substrates containing pharmaceutical or other skin care active, moisturizing substrates, sunscreen substrates, chronic skin benefit agent substrates (e.g., vitamin-containing substrates, alpha-hydroxy acid-containing substrates, etc.), deodorizing substrates, fragrance-containing substrates, and so forth.

(102) The dissolvable solid structure can be a personal care product, alternatively hair care product, alternatively rinse-off hair care product, alternatively rinse-off hair care product containing non-sulfate surfactant.

(103) Described herein is an article of commerce comprising one or more dissolvable solid structures described herein, and a communication directing a consumer to dissolve the Structure and apply the dissolved mixture to hair, hair follicles, skin, teeth, the oral cavity, to achieve a benefit to the target consumer substrate, a rapidly lathering foam, a rapidly rinsing foam, a clean rinsing foam, and combinations thereof. The communication may be printed material attached directly or indirectly to packaging that contains the dissolvable solid structure or on the dissolvable solid structure itself. Alternatively, the communication may be an electronic or a broadcast message that is associated with the article of manufacture. Alternatively, the communication may describe at least one possible use, capability, distinguishing feature and/or property of the article of manufacture.

(104) Test Methods

(105) Basis Weight Measurement

(106) In general, basis weight of a material or article (including the dissolvable solid structure) is measured by first cutting the sample to a known area, using a die cutter or equivalent, then measuring & recording the weight of the sample on a top-loading balance with a minimum resolution of 0.01 g, then finally by calculating the basis weight as follows:

Basis Weight (g/m.sup.2)=weight of basis weight pad (g)

$$(107) \text{BasisWeight}\left(\frac{g}{m^2}\right) = \frac{\text{Weightofpad}(g) \times 10,000 \frac{cm^2}{m^2}}{\text{areaofpad}(cm^2)}$$

(108) Suitable pad sample sizes for basis weight determination are >10 cm.sup.2 and should be cut with a precision die cutter having the desired geometry. If the dissolvable solid structure to be measured is smaller than 10 cm.sup.2, a smaller sampling area can be used for basis weight determination with the appropriate changes to calculation.

(109) In the present examples, basis weight was calculated based on the full dissolvable solid structure having a known area of 17.28 cm.sup.2. Thus, the basis weight calculation becomes:

$$(110) \text{BasisWeight}\left(\frac{g}{m^2}\right) = \frac{\text{Weightofpad}(g) \times 10,000 \frac{cm^2}{m^2}}{17.28 cm^2}$$

Hand Dissolution Test Method

Materials Needed:

(111) Dissolvable solid structures to be tested: 3-5 dissolvable solid structure s (finished product samples) are tested so that an average of the number of strokes for each if the individual dissolvable solid structure samples is calculated and recorded as the Average Hand Dissolution value for the dissolvable solid structure. For this method, the entire consumer saleable or consumer use dissolvable solid structure is tested. If the entire consumer saleable or consumer use dissolvable solid structure has a footprint greater than 50 cm.sup.2, then first cut the dissolvable solid structure to have a footprint of 50 cm.sup.2.

(112) Nitrile Gloves

(113) 10 cc syringe Plastic Weigh boat (~3 in×3 in) 100 mL Glass beaker Water (City of Cincinnati Water or equivalent having the following properties: Total Hardness=155 mg/L as CaCO₂; Calcium content=33.2 mg/L; Magnesium content=17.5 mg/L; Phosphate content=0.0462 mg/L) Water used is 7 gpg hardness and 40° C.+/-5° C.

Protocol: 1. Add 180 mL of water to glass beaker. Add 300-400 ml of water to glass beaker. 2. Heat water in beaker until water is at a temperature of 40° C. +/- 5° C. 3. Transfer 10 mL of the water from the beaker into the weigh boat via the syringe. 4. Within 10 seconds of transferring the water to the weigh boat, place dissolvable solid structure sample in palm of gloved hand (hand in cupped position in non-dominant hand to hold dissolvable solid structure sample). 5. Using dominant hand, add water quickly from the weigh boat to the dissolvable solid structure sample and allow to immediately wet for a period of 5-10 seconds. 6. Rub with opposite dominant hand (also gloved) in 2 rapid circular strokes. 7. Visually examine the dissolvable solid structure sample in hand after the 2 strokes. If dissolvable solid structure sample is completely dissolved, record number of strokes=2 Dissolution Strokes. If not completely dissolved, rub remaining dissolvable solid structure sample for 2 more circular strokes (4 total) and observe degree of dissolution. If the dissolvable solid structure sample contains no solid pieces after the 2 additional strokes, record number of strokes=4 Dissolution Strokes. If after the 4 strokes total, the dissolvable solid structure sample still contains solid pieces of un-dissolved dissolvable solid structure sample, continue rubbing remaining dissolvable solid structure sample in additional 2 circular strokes and check if there are any remaining solid pieces of dissolvable solid structure sample after each additional 2 strokes until dissolvable solid structure sample is completely dissolved or until reaching a total of 30 strokes, whichever comes first. Record the total number of strokes. Record 30 Dissolution Strokes even if solid dissolvable solid structure sample pieces remain after the maximum of 30 strokes. 8. Repeat this process for each of the additional 4 dissolvable solid structure samples. 9. Calculate the arithmetic mean of the recorded values of Dissolution Strokes for the 5 individual dissolvable solid structure samples and record as the Average Hand Dissolution Value for the dissolvable solid structure. The Average Hand Dissolution Value is reported to the nearest single Dissolution Stroke unit.

Fibrous Structures—Fiber Diameter

(114) For fibrous Structures, the diameter of dissolvable fibers in a sample of a web is determined by using a Scanning Electron Microscope (SEM) or an Optical Microscope and image analysis software. A magnification of 200 to 10,000 times is chosen such that the fibers are suitably enlarged for measurement. When using the SEM, the samples are sputtered with gold or a palladium compound to avoid electric charging and vibrations of the fibers in the electron beam. A manual procedure for determining the fiber diameters is used from the image (on monitor screen) taken with the SEM or the optical microscope. Using a mouse and a cursor tool, the edge of a randomly selected fiber is sought and then measured across its width (i.e., perpendicular to fiber direction at that point) to the other edge of the fiber. A scaled and calibrated image analysis tool provides the scaling to get actual reading in microns (μm). Several fibers are thus randomly selected across the sample of the web using the SEM or the optical microscope. At least two specimens from the web (or web inside a product) are cut and tested in this manner. Altogether at least 100 such measurements are made and then all data are recorded for statistical analysis. The recorded data are used to calculate average (mean) of the fiber diameters, standard deviation of the fiber diameters, and median of the fiber diameters. Another useful statistic is the calculation of the amount of the population of fibers that is below a certain upper limit. To determine this statistic, the software is programmed to count how many results of the fiber diameters are below an upper limit and that count (divided by total number of data and multiplied by 100%) is reported in percent as percent below the upper limit, such as percent below 1 micron diameter or %-submicron, for example. We denote the measured diameter (in microns) of an individual circular fiber as d_i . (115) In case the fibers have non-circular cross-sections, the measurement of the fiber diameter is determined as and set equal to the hydraulic diameter which is four times the cross-sectional area of the fiber divided by the perimeter of the cross of the fiber (outer perimeter in case of hollow fibers). The number-average diameter, alternatively average diameter is calculated as, $d_{\text{sub.num}}$

$$(116) \frac{\sum_{i=1}^n \text{Math. } d_i}{n}$$

Combinations 1. A dissolvable solid structure, wherein the dissolvable solid structure is formed by a homogeneous mixture comprising, by weight of the structure: from about 1% to about 10% of a first polymeric structurant having a molecular weight of from about 1,000,000 g/mol to about 4,000,000 g/mol; from about 10% to about 20% of a second polymeric structurant having a molecular weight of from about 50,000 g/mol to about 500,000 g/mol; from about 10 wt % to about 85 wt % of one or more high melting point fatty material having a carbon chain length C12-C22 or mixtures thereof, wherein the melting point is above 25° C.; and from about 1 wt % to about 60 wt % of a cationic surfactant. 2. The dissolvable solid structure of the preceding feature, wherein the first and second polymeric structurants are selected from the group consisting of: polyacrylic acid and its copolymers, polyvinylpyrrolidone and its copolymers, polyacrylamide and its copolymers, polyvinylmethyl ether, polyethyleneimine, polymethacrylic acid, poly N-isopropyl acrylamide, poly N—N-dimethylacrylamide, polyvinylloxazolidone, a poly(vinyl oxazoline), poly 2-ethyl-2-oxazoline, polyvinylcaprolactam, polystyrene sulfonate, polyvinyl formamide, polyvinyl amine, poly alkylated vinyl pyrrolidone, poly vinyl valerolactam, polyvinyl imidazole, polyacrylic acid, polymethacrylate, polyacrylamide, polymethacrylamide, polydimethacrylamide, polyalkylaminomethacrylate, and polyalkylaminomethacrylamide, and copolymer combinations thereof. 3. The dissolvable solid structure of any of the preceding features, wherein the first and second polymeric structurants are selected from the group consisting of: polyvinylpyrrolidone and its copolymer, and mixtures thereof, preferably wherein the first and second polymeric structurants are polyvinylpyrrolidone. 4. The dissolvable solid structure of any of the preceding features, wherein the first polymeric structurant has a molecular weight of from about 1,500,000 g/mol to about 3,500,000 g/mol. 5. The dissolvable solid structure of any of the preceding features, wherein the second polymeric structurant has a molecular weight of from about 65,000 g/mol to about 395,000 g/mol. 6. The dissolvable solid structure of any of the preceding features, wherein the weight ratio between the first and second polymeric structurants is from about 1:1 to about 1:100, preferably from about 1:1 to about 1:50, more preferably from about 1:1 to about 1:10. 7. The dissolvable solid structure of any of the preceding features, wherein the weight ratio between the first and second polymeric structurants is in a range such that the average molecular weight of the first and second polymeric structurants is from about 300,000 g/mol to about 3,000,000 g/mol, preferably from about 350,000 g/mol to about 2,000,000 g/mol, more preferably from about 400,000 g/mol to about 1,000,000 g/mol, and even more preferably from about 500,000 g/mol to about 1,000,000 g/mol. 8. The dissolvable solid structure of any of the preceding features, wherein a total amount of the first and second polymeric structurants is from about 11% to about 25% by weight of the structure, preferably from about 13% to about 20% by weight of the structure. 9. The dissolvable solid structure of any of the preceding features, wherein the dissolvable solid structure is a personal care product, preferably hair care product, more preferably rinse-off hair care product, still more preferably rinse-off hair care product containing non-sulfate surfactant. 10. The dissolvable solid structure of any of the preceding features, comprising from about 10 wt % to about 50 wt % of cationic surfactant by weight of the structure, preferably from about 20 wt % to about 40 wt % of cationic surfactant. 11. The dissolvable solid structure of any of the preceding features, wherein the Dissolvable Solid Structure dissolves in less than 30 strokes of the Hand Dissolution Method, preferably wherein the Dissolvable Solid Structure dissolves in less than 20 strokes of the Hand Dissolution Method, and even more preferably wherein the Dissolvable Solid Structure dissolves in less than 15 strokes of the Hand Dissolution Method.

NON-LIMITING EXAMPLES

(117) The compositions illustrated in the following Examples illustrate specific embodiments of the composition, but are not intended to be limiting thereof. Other modifications can be undertaken by the skilled artisan without departing from the spirit and scope of this invention. These exemplified embodiments of the composition as described herein provide enhanced conditioning benefits to the hair.

(118) All exemplified amounts are listed as weight percents and exclude minor materials such as diluents, preservatives, color solutions, imagery ingredients, botanicals, and so forth, unless otherwise specified. All percentages are based on weight unless otherwise specified.

(119) TABLE-US-00001 Dissolvable solid structure-Hair conditioner Examples Raw Material Ex 1 Ex 2 Ex 3 Ex 4 Ex 5 CEx. i CEx. ii Behentrimonium 21 21.5 21.7 21.8 — — 25 Methosulfate Behentrimonium — — — 14 20 — Chloride Stearyl Alcohol 37 38 38.2 38.4 41.3 35 40 1-Hexadecanol 15 15.5 15.6 15.7 20.7 15 20 Lauroyl Methyl 10 9.5 7.8 7.8 7 10 10 Glucamide Polyvinyl pyrrolidone 10 9.3 13.3 10.7 10 18 — (PVP K30 from Ashland) having MW of 66,800 Polyvinyl pyrrolidone — — — 1.6 — — — (PVP K60 from Jarchem) having MW of 396,000 Polyvinyl pyrrolidone 5.5 5 3.3 — 5.5 5 — (PVP K90 from Ashland) having MW of 1,570,000 Polyvinyl pyrrolidone 1.5 1.2 — 4 1.5 — 4 (PVP

K120 from Ashland) having MW of 3,470,000 Combined average 853,412 815,177 365,629 934,255 853,412 66,800 3,470,000 MW of PVPs Strength during Pass Pass Pass Pass Pass Pass Fail manufacturing *1 Usage feel *2 Pass Pass Pass Pass Pass Fail Pass *1 Mechanical strength during any suitable method to manufacturing the dissolvable solid structure which is a fibrous structure formed by a plurality of fibrous elements. For example, the fibrous elements are formed by meltblowing and/or spunbonding, by using a fibrous element-forming composition comprising the above ingredients in the table. The levels of the ingredients in such fibrous element-forming composition are adjusted such that the ingredients have the levels defined above in the dissolvable solid structure. *2 Usage feel is assessed as follows: Samples provided to 10 panelists who rate the product based on overall hair feel compared to a control conditioner (CEx.ii). "Pass" means that 6 or more panelists answer that the overall hair feel is parity to or superior than the control conditioner.

(120) The dissolvable solid structures of Ex.1 to Ex. 5 are examples of the present invention These examples of the present invention provide an adequate strength during manufacturing, while not deteriorating usage feel when used and/or applied by consumers.

(121) The dissolvable solid structures of CEx.i and CEx.ii are comparative examples of the present invention. In CEx.i, the mechanical strength of the fibrous element-forming composition during manufacturing is sufficient, however, its overall hair feel is inferior than the control conditioner. In CEx.ii, the mechanical strength of the fibrous element-forming composition during manufacturing too low while it exhibits acceptable hair feel.

(122) The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as "40 mm" is intended to mean "about 40 mm."

(123) Every document cited herein, including any cross referenced or related patent or application, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

(124) While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

Claims

1. A dissolvable solid structure, wherein the dissolvable solid structure is formed by a homogeneous mixture comprising, by weight of the structure: polymeric structurants consisting of: from about 1% to about 10% of one or more first polymeric structurants wherein the structurants comprise a molecular weight of from about 1,500,000 g/mol to about 3,500,000 g/mol; from about 10% to about 20% of or more second polymeric structurants wherein the structurants comprise a molecular weight of from about 65,000 g/mol to about 396,000 g/mol; wherein the ratio of the first polymeric structurants to the second polymeric structurants is less than 1:1; wherein the first and second polymeric structurants are polyvinylpyrrolidone, its copolymer, or a mixture thereof; from about 10 wt % to about 85 wt % of one or more high melting point fatty material comprising a carbon chain length C12-C22 or mixtures thereof and a melting point above 25° C.; and from about 1 wt % to about 60 wt % of a cationic surfactant; wherein the combined average molecular weight of the first and second polymeric structurants is from about 350,000 g/mol to about 1,000,000 g/mol; wherein the dissolvable structure is a hair conditioner.
 2. The dissolvable solid structure of claim 1 wherein the first and second polymeric structurants are polyvinylpyrrolidone.
 3. The dissolvable solid structure of claim 1, wherein a total amount of the first and second polymeric structurants is from about 11% to about 25% by weight of the structure.
 4. The dissolvable solid structure of claim 3, wherein a total amount of the first and second polymeric structurants is from about 13% to about 20% by weight of the structure.
 5. The dissolvable solid structure of claim 1, comprising from about 10 wt % to about 50 wt % of cationic surfactant by weight of the structure.
 6. The dissolvable solid structure of claim 5, comprising from about 20 wt % to about 40 wt % of cationic surfactant by weight of the structure.
 7. The dissolvable solid structure of claim 1, wherein the dissolvable solid structure dissolves in less than 20 strokes, according to a Hand Dissolution Method described herein.
 8. The dissolvable solid structure of claim 7, wherein the dissolvable solid structure dissolves in less than 15 strokes, according to the Hand Dissolution Method described herein.
 9. The dissolvable solid structure of claim 1, wherein the cationic surfactant is chosen from behentrimonium methosulfate, behentrimonium chloride, or mixtures thereof.
 10. The dissolvable solid structure of claim 1, wherein the fatty material is stearyl alcohol.
 11. The dissolvable solid structure of claim 1, wherein the structurants consist of two first polymeric structurants and one second polymeric structurant.
 12. The dissolvable solid structure of claim 11, wherein the two first polymeric structurants have a molecular weight of about 1,570,000 and about 3,470,000 and the second polymeric structurant has a molecular weight of about 66,800.
 13. The dissolvable solid structure of claim 1, wherein the structurants consist of one first polymeric structurant and two second polymeric structurants.
 14. The dissolvable solid structure of claim 13, wherein the first polymeric structurant comprises a molecular weight of about 3,470,000 and the two second polymeric structurant comprises a molecular weight of about 66,800 and about 396,000.
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